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BATTERY CELL WITH IMPROVED WELDABILITY AND APPARATUS FOR PROCESSING THE SAME

Abstract

A battery cell of a secondary battery can include an electrode assembly; a cell case accommodating the electrode assembly; and a pair of electrode leads protruding outwards from the cell case, wherein the pair of electrode leads include a first electrode lead including a material that is the same as a material of a plate configured to connect the battery cell to another battery cell in series or parallel; and wherein the pair of electrode leads include a second electrode lead including a material that is different from the material of the first electrode lead.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Divisional of U.S. patent application Ser. No. 17/604,813, filed on Oct. 19, 2021, which is the National Phase of PCT International Application No. PCT/KR2020/018261 filed on Dec. 14, 2020, which claims priority under 35 U.S.C. § 119(a) to Korean Patent Application No. 10-2020-0010843, filed in the Republic of Korea on Jan. 30, 2020, the entire contents of which is hereby expressly incorporated by reference into the present application.

BACKGROUND OF THE DISCLOSURE

Field of Disclosure

[0002] The present invention relates to a battery cell with improved weldability and an apparatus for processing the same, and more particularly to a battery cell with improved weldability configured such that a negative electrode lead is provided with an embossed portion, whereby weld defects are minimized, and such that the embossed portion can be formed in the negative electrode lead simultaneously with cutting of leads, whereby the battery cell is easily processed, and an apparatus for processing the same.

Discussion of the Related Art

[0003] With technological development of mobile devices, such as mobile phones, laptop computers, camcorders, and digital cameras, and an increase in demand therefor, research on secondary batteries, which are capable of being charged and discharged, has been actively conducted. In addition, secondary batteries, which are energy sources substituting for fossil fuels causing air pollution, have been applied to an electric vehicle (EV), a hybrid electric vehicle (HEV), and a plug-in hybrid electric vehicle (P-HEV), and therefore there is an increasing necessity for development of secondary batteries.

[0004] There are a nickel-cadmium battery, a nickel-hydride battery, a nickel-zinc battery, and a lithium secondary battery as currently commercialized secondary batteries. Thereamong, the lithium secondary battery is in the spotlight, since the lithium secondary battery has little memory effect, whereby the lithium secondary battery is capable of being freely charged and discharged, has a very low self-discharge rate, and has high energy density, compared to the nickel-based secondary batteries.

[0005] In the case in which the above secondary battery is used in a device that requires large capacity and high voltage, such as an electric vehicle, a separate metal plate is necessary in order to connect a plurality of battery cells to each other in series or in parallel. That is, a plurality of electrode leads is coupled to each other, or an electrode lead and a metal plate are coupled to each

other, using a known joining method, such as welding. At this time, however, weld defects occur due to various causes, whereby manufacturing costs are increased.

[0006] As shown in FIG. 1, which is a perspective view of a conventional battery cell, the conventional battery cell **10** includes an electrode assembly, a pair of electrode leads **12** welded to a plurality of electrode tabs **11** extending from the electrode assembly, and a case **13** configured to receive the electrode assembly. However, since each of the electrode leads **12** has a simple flat shape, as described above, the electrode leads may not be properly brought into tight contact with an object to be welded, whereby there is a high possibility of occurrence of weld defects.

[0007] Meanwhile, a battery cell undergoes a process of cutting leads thereof so as to satisfy prescribed standards. At this time, the leads may be bent and thus may not be cut at predetermined positions thereof, or cutters configured to cut the leads must be frequently replaced.

[0008] FIG. 2 is a side view of a conventional battery cell processing apparatus. As shown in FIG. 2, the conventional processing apparatus is configured such that a seating jig **22** is located on the upper surface of a base **21** and such that a cutting body **23**, a pneumatic member **24**, a cutting member **25**, and an upper cutting knife **26** are provided adjacent thereto.

[0009] At the time of processing a battery cell, therefore, a cell case **13** is mounted on the seating jig **22** such that a lead **12** is exposed in the direction in which the cutting body **23** is located, the upper cutting knife **26** is moved downwards, and the lead **12** is cut at the point at which a lower cutting knife **27** is located.

[0010] Since the upper cutting knife **26** and the side surface of the seating jig **22** are greatly spaced apart from each other, however, the portion of the lead **12** located on the right side of the upper cutting knife **26** is bent downwards, whereas the portion of the lead **12** located on the left side of the lower cutting knife **27** is bent upwards. In many cases, therefore, the cut surface may not be vertical.

[0011] Furthermore, the upper cutting knife and the lower cutting knife, which form a pair of cutting knives, are provided, whereby the apparatus is complicated. Also, in the case in which the cutting knives become dull, both of the cutting knives must be replaced, whereby maintenance expenses are increased.

PRIOR ART DOCUMENT

[0012] (Patent Document 1) Korean Patent Application Publication No. 2006-0011312

SUMMARY OF THE DISCLOSURE

[0013] The present invention has been made in view of the above problems, and it is an object of the present invention to provide a battery cell configured such that weld quality of the battery cell can be improved at the time of resistance welding.

[0014] It is another object of the present invention to provide a battery cell processing apparatus capable of reducing maintenance expenses while accurately cutting leads of a battery cell.

[0015] In order to accomplish the above objects, a battery cell according to the present invention is configured such that an electrode assembly is received in a cell case (**110**) and a pair of electrode leads constituted by a negative electrode lead (**120**) and a positive electrode lead (**130**) protrudes outwards from the cell case (**110**), wherein the negative electrode lead (**120**) is made of a nickel material and is provided with one or more embossed portions (**121**), and the positive electrode lead (**130**) is made of an aluminum material and is provided with no embossed portion.

[0016] In addition, a battery cell processing apparatus according to the present invention includes a base portion (**210**) having a seating die (**211**), configured to support a cell case (**110**) and a pair of leads, provided on the upper surface thereof, a vertical fixing portion (**220**) fixed to a predetermined region of one side of the base portion (**210**) so as to be perpendicular to the base portion, a block portion (**230**) connected to the vertical fixing portion (**220**) so as to be movable upwards and downwards relative to the vertical fixing portion, a pair of cutters (**250**) located at the bottom surface of the block portion (**230**), the pair of cutters being configured to cut a negative electrode lead (**120**) and a positive electrode lead (**130**), the pair of cutters being spaced apart from

each other by a predetermined distance, and a pair of support portions (270) located at the bottom surface of the block portion (230), the pair of support portions being located parallel to the pair of cutters (250) in a horizontal direction, wherein one of the pair of support portions (270) is provided on the bottom surface thereof with protrusions (271(a)) configured to form embossed portions (121) in the negative electrode lead (120).

[0017] Also, the battery cell processing apparatus according to the present invention may further include a second connection portion (260) configured to connect the block portion (230) and the pair of cutters (250) to each other and a third connection portion (280) configured to connect the block portion (230) and the pair of support portions (270) to each other, wherein the third connection portion (280) may be constituted by an elastic member capable of being expanded and contracted in a vertical direction, and the second connection portion (260) may not be capable of being expanded and contracted in the vertical direction.

[0018] Also, in the battery cell processing apparatus according to the present invention, the third connection portion (280) may include a spring.

[0019] Also, in the battery cell processing apparatus according to the present invention, the pair of cutters (250) may include a first cutter (251) and a second cutter (252), the pair of support portions (270) may include a first support portion (271) and a second support portion (272), and the first cutter (251) and the first support portion (271) may be located adjacent to each other while the second cutter (252) and the second support portion (272) may be located adjacent to each other.

[0020] Also, in the battery cell processing apparatus according to the present invention, each of the cutters (250) may have a polyhedral shape.

[0021] Also, in the battery cell processing apparatus according to the present invention, each of the cutters (250) has a hexahedral shape.

[0022] Also, in the battery cell processing apparatus according to the present invention, each of the cutters (250) may include a front portion (251(a)), a rear portion (251(b)), a bottom portion (251(c)) configured to connect lower parts of the front portion (251(a)) and the rear portion (251(b)) to each other, a pair of edge portions (251(d)) formed at the position at which the front portion (251(a)) and the bottom portion (251(c)) join each other and the position at which the rear portion (251(b)) and the bottom portion (251(c)) join each other, and a pair of side portions (251(e)) configured to connect side parts of the front portion (251(a)) and the rear portion (251(b)) to each other, and each of the pair of side portions (251(e)) may be further provided with an extension portion (251(f)) extending a predetermined length in a downward direction.

[0023] Also, in the battery cell processing apparatus according to the present invention, each of the edge portions (251(d)) may form a right angle.

[0024] Also, in the battery cell processing apparatus according to the present invention, one side surface of each of the cutters (250) may be moved along one side surface of the seating die (211) in the vertical direction, and each of the support portions (270) may be moved above the upper surface of the seating die (211) in the vertical direction.

[0025] Also, in the battery cell processing apparatus according to the present invention, the distance between the one side surface of the seating die (211) and the one side surface of each of the cutters (250) may be greater than 0 mm to less than 0.05 mm.

[0026] In addition, a battery cell processing method according to the present invention includes a first step of placing a battery cell to be processed on the upper surface of a seating die (211) of a base portion (210), a second step of moving the seating die (211) forwards, a third step of moving a block portion (230) downwards in the vertical direction to cut a negative electrode lead (120) and a positive electrode lead (130) to a predetermined length and at the same time to form one or more embossed portions (121) in the negative electrode lead (120), and a fourth step of separating the processed battery cell from the seating die (211), the steps being repeatedly performed, wherein, after a predetermined number of battery cells are processed, cutters (250) are rotated 180 degrees in a clockwise direction or in a counterclockwise direction, and the steps are repeatedly performed.

[0027] A battery cell according to the present invention has an advantage in that a negative electrode lead of the battery cell is provided with one or more embossed portions, whereby weld quality of the battery cell is improved at the time of resistance welding with a metal plate.

[0028] In addition, a battery cell processing apparatus according to the present invention has an advantage in that a cutter configured to cut a lead and a support portion configured to form an embossed portion and to support the lead are simultaneously moved upwards and downwards by driving of a block portion, whereby processing accuracy is high and processing time is reduced.

[0029] Furthermore, the cutter according to the present invention has a tetragonal shape, whereby durability of the cutter is excellent, and a pair of opposite edges of the cutter can be used as the edges of a knife, which is economical.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1 is a perspective view of a conventional battery cell.

[0031] FIG. 2 is a side view of a conventional battery cell processing apparatus.

[0032] FIG. 3 is a perspective view of a battery cell according to a preferred embodiment of the present invention.

[0033] FIG. 4 is a side view of a battery cell processing apparatus according to a preferred embodiment of the present invention.

[0034] FIG. 5 is a perspective view of the battery cell processing apparatus according to the preferred embodiment of the present invention.

[0035] FIG. 6 is a perspective view of a cutter according to a preferred embodiment of the present invention (FIG. 6(a) is a front perspective view and FIG. 6(b) is a rear perspective view).

[0036] FIG. 7 is a perspective view of support portions according to a preferred embodiment of the present invention (FIG. 7(a) is a perspective view of a first support portion and FIG. 7(b) is a perspective view of a second support portion).

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0037] In the present application, it should be understood that the terms “comprises,” “has,” “includes,” etc. specify the presence of stated features, numbers, steps, operations, elements, components, or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, steps, operations, elements, components, or combinations thereof.

[0038] In addition, the same reference numbers will be used throughout the drawings to refer to parts that perform similar functions or operations. In the case in which one part is said to be connected to another part in the specification, not only may the one part be directly connected to the other part, but also, the one part may be indirectly connected to the other part via a further part. In addition, that a certain element is included does not mean that other elements are excluded, but means that such elements may be further included unless mentioned otherwise.

[0039] Hereinafter, a battery cell with improved weldability, a battery cell processing apparatus, and a battery cell processing method according to the present invention will be described with reference to the accompanying drawings.

[0040] FIG. 3 is a perspective view of a battery cell according to a preferred embodiment of the present invention. As shown in FIG. 3, the battery cell according to the present invention includes a cell case **110**, an electrode assembly (not shown), and a pair of leads.

[0041] The cell case **110** is generally configured to have a laminate sheet structure including an inner layer, a metal layer, and an outer layer. The inner layer is disposed in direct contact with the electrode assembly, and therefore the inner layer must exhibit high insulation properties and high resistance to an electrolytic solution. In addition, the inner layer must exhibit high sealability in order to hermetically seal the cell case from the outside, i.e. a thermally-bonded sealed portion

between inner layers must exhibit excellent thermal bonding strength. The inner layer may be made of a material selected from among a polyolefin-based resin, such as polypropylene, polyethylene, polyethylene-acrylic acid, or polybutylene, a polyurethane resin, and a polyimide resin, which exhibit excellent chemical resistance and high sealability. However, the present invention is not limited thereto, and polypropylene, which exhibits excellent mechanical-physical properties, such as tensile strength, rigidity, surface hardness, and resistance to impact strength, and excellent chemical resistance, is the most preferably used.

[0042] The metal layer, which is disposed so as to abut the inner layer, corresponds to a barrier layer configured to prevent moisture or various kinds of gas from permeating into the battery from the outside. Aluminum foil, which is light and easily formable, may be used as a preferred material for the metal layer.

[0043] The outer layer is provided on the other surface of the metal layer. The outer layer may be made of a heat-resistant polymer that exhibits excellent tensile strength, resistance to moisture permeation, and resistance to air transmission such that the outer layer exhibits high heat resistance and chemical resistance while protecting the electrode assembly. As an example, the outer layer may be made of nylon or polyethylene terephthalate. However, the present invention is not limited thereto.

[0044] The electrode assembly constituting the battery cell **100** may be a jelly-roll type electrode assembly, which is configured to have a structure in which a long sheet type positive electrode and a long sheet type negative electrode are wound in the state in which a separator is interposed between the positive electrode and the negative electrode, a stacked type electrode assembly including unit cells, each of which is configured to have a structure in which a rectangular positive electrode and a rectangular negative electrode are stacked in the state in which a separator is interposed therebetween, a stacked and folded type electrode assembly, which is configured to have a structure in which unit cells are wound using a long separation film, or a laminated and stacked type electrode assembly, which is configured to have a structure in which unit cells are stacked in the state in which a separator is interposed therebetween and are then attached to each other. However, the present invention is not limited thereto.

[0045] The pair of electrode leads includes a negative electrode lead **120** and a positive electrode lead **130**. The leads may be exposed out of the cell case **110** in the state in which a negative electrode tab and a positive electrode tab of the electrode assembly are electrically connected respectively to the leads. Alternatively, the negative electrode lead **120** and the positive electrode lead **130** may be directly connected to the electrode assembly without the negative electrode tab and the positive electrode tab.

[0046] Here, it is preferable that the negative electrode lead **120** be made of a nickel material and be provided with one or more embossed portions **121**, more preferably two or more embossed portions. It is more preferable that the positive electrode lead **130** be made of an aluminum material and be formed in a flat shape having no embossed portion.

[0047] The battery cell **100** must be joined to a metal plate, such as a nickel plate, for parallel connection or series connection. In the case in which the embossed portions **121** are provided in the negative electrode lead **120** made of the nickel material, welding defects due to weak-welding or non-welding may be minimized at the time of resistance welding.

[0048] On the other hand, since the melting point of the positive electrode lead **130** is lower than the melting point of nickel, the positive electrode lead may not be properly joined but may be ruptured at the time of resistance welding. At the time of resistance welding between the nickel plate and the aluminum positive electrode lead, therefore, it is advantageous that no embossed portion is formed in the positive electrode lead in order to secure sufficient contact area.

[0049] FIG. **4** is a side view of a battery cell processing apparatus according to a preferred embodiment of the present invention, and FIG. **5** is a perspective view of the battery cell processing apparatus according to the preferred embodiment of the present invention.

[0050] Referring to FIGS. 4 and 5, the battery cell processing apparatus **200** according to the preferred embodiment of the present invention includes a base portion **210**, a vertical fixing portion **220**, a block portion **230**, a first connection portion **240**, a pair of cutters **250**, a pair of second connection portions **260**, a pair of support portions **270**, and a pair of third connection portions **280**. [0051] The base portion **210** may be configured to have the shape of an approximately quadrangular plate having a flat upper surface, and a seating die **211** located on the upper surface may be mounted so as to be slidable forwards or rearwards, which is an X-axis direction. Here, it is preferable that one side surface of the seating die **211**, on which the negative electrode lead **120** is seated and from which the negative electrode lead protrudes, be perpendicular to the negative electrode lead.

[0052] The vertical fixing portion **220** located at one side of the base portion **210**, specifically, the vertical fixing portion **220** fixed to the rear of the base portion **210** so as to be perpendicular to the base portion, is configured to hold the pair of cutters **250** and the pair of support portions **270** connected to the block portion **230** so as to be movable upwards and downwards in the state of being spaced apart from the seating die **211** by a predetermined distance. That is, the block portion **230** is movable upwards and downwards (in a Y-axis direction) through the intervention of the first connection portion **240**, one side of which is connected to the vertical fixing portion **220** and the other side of which is connected to the block portion **230**.

[0053] Of course, it is obvious that a separate pneumatic means (not shown) is mounted to the first connection portion **240** in order to move the block portion **230** upwards and downwards.

[0054] Meanwhile, it is more preferable that the vertical fixing portion **220** be constituted by two pillars or a single pillar having a large width in order to secure stability thereof, although the vertical fixing portion is configured to have a single pillar shape having a small width.

[0055] The block portion **230**, which is supported by the vertical fixing portion **220** so as to be movable upwards and downwards, may have a predetermined shape, such as a hexahedral shape. Here, the block portion **230** may be constituted by two parts. However, it is more preferable that a single block portion be provided in order to simultaneously move the pair of cutters **250** and the pair of support portions **270** upwards and downwards by one manipulation.

[0056] The pair of cutters **250** and the pair of support portions **270** are located under the block portion **230**. A first cutter **251** and a second cutter **252**, which constitute the pair of cutters **250**, are configured to cut the negative electrode lead **120** and the positive electrode lead **130** to a predetermined length, respectively.

[0057] The pair of support portions **270**, which includes a first support portion **271** and a second support portion **272**, is configured to prevent the leads from being bent at the time of cutting, thereby minimizing product defects. In addition, the pair of support portions is configured to form the embossed portions **121** in the negative electrode lead **120** simultaneously with cutting.

[0058] FIG. 6 is a perspective view of a cutter according to a preferred embodiment of the present invention (FIG. 6(a) is a front perspective view and FIG. 6(b) is a rear perspective view), and FIG. 7 is a perspective view of support portions according to a preferred embodiment of the present invention (FIG. 7(a) is a perspective view of a first support portion and FIG. 7(b) is a perspective view of a second support portion).

[0059] The first cutter **251** and the second cutter **252** according to the present invention are identical in construction to each other. Consequently, only the first cutter **251** will be described in detail with reference to FIG. 6.

[0060] The first cutter **251** includes a front portion **251(a)**, a rear portion **251(b)**, a bottom portion **251(c)**, a pair of edge portions **251(d)**, and a pair of side portions **251(e)**.

[0061] In particular, the bottom portion **251(c)** connects lower parts of the front portion **251(a)** and the rear portion **251(b)** to each other, and the pair of edge portions **251(d)**, which is formed at the position at which the front portion **251(a)** and the bottom portion **251(c)** join each other and at the position at which the rear portion **251(b)** and the bottom portion **251(c)** join each other, serves as

the edge of a knife configured to cut the lead.

[0062] Here, it is preferable that the angles of the edge portions **251(d)**, i.e. the angle between the front portion **251(a)** and the bottom portion **251(c)** and the angle between the rear portion **251(b)** and the bottom portion **251(c)**, be a right angle. This configuration is provided to cut each of the negative electrode lead **120** and the positive electrode lead **130** so as to have a vertical cut surface.

[0063] Each of the pair of side portions **251(e)** may be further provided with an extension portion **251(f)** extending a predetermined length in a downward direction. This configuration is provided to allow the cutter **251** to move along slit grooves formed in the seating die **211** at the time of upward and downward movement thereof to thus accurately guide sliding of the cutter **251** in a vertical direction without being twisted leftwards and rightwards.

[0064] Of course, the first cutter **251** may have a polyhedral shape with the extension portions **251(f)** omitted, for example, a hexahedral shape.

[0065] As shown in FIG. 7, the support portions **270** include a first support portion **271** configured to support the negative electrode lead **120** and at the same time to form the embossed portions **121** and a second support portion **272** configured to support the positive electrode lead **130**. The bottom surface of each of the support portions **270** is flat so as to support the upper surface of a corresponding one of the leads in the state of being in entirely tight contact therewith.

[0066] Of course, the first support portion **271** is provided on the bottom surface thereof with protrusions **271a** equal in number to embossed portions **121** to be formed.

[0067] Referring back to FIGS. 4 and 5, the cutters **250** and the support portions **270** having the constructions described above are disposed as follows. The support portions **270** are located at positions relatively close to the cell case **110**, and the cutters **250** are located at positions relatively distant from the cell case. In addition, the support portions **270** and the cutters **250** are located so as to be adjacent to each other.

[0068] Specifically, the first cutter **251** and the first support portion **271** constitute one assembly, and the second cutter **252** and the second support portion **272** constitute another assembly. The distance between each of the support portions and a corresponding one of the cutters may be greater than 0 mm to less than 0.1 mm. In the case in which the support portion and the cutter are disposed in tight contact with each other, the side surface of the support portion and/or the cutter may be worn by friction, and great driving force is necessary to move the support portion and the cutter upwards and downwards. In the case in which the support portion and the cutter are disposed so as to be excessively spaced apart from each other, on the other hand, it is not possible to inhibit bending of the lead, whereby it is not possible to accurately cut the lead.

[0069] Meanwhile, the cutters **250** are connected to the block portion **230** via the second connection portions **260**, and the support portions **270** are connected to the block portion **230** via the third connection portions **280**. Specifically, the first cutter **251** and the second cutter **252** are connected to a 2' connection portion **261** and a 2'' connection portion **262**, respectively, and the first support portion **271** and the second support portion **272** are connected to a 3' connection portion **281** and a 3'' connection portion **282**, respectively.

[0070] Here, each of the third connection portions **280** connected to the support portions **270**, i.e. each of 3' connection portion **281** and the 3'' connection portion **282**, is constituted by an elastic member capable of being expanded and contracted in the vertical direction, such as a spring, whereas each of the 2' connection portion **261** and the 2'' connection portion **262** connected to the cutters **250** is configured in the form of a bar having no length change.

[0071] Of course, when force of expansion and contraction is insufficient by the provision of only the elastic members, such as the springs, air cylinders may be further provided between the block portion **230** and the third connection portions **280** and/or between the third connection portions **280** and the support portions **270**, although not shown in the figures.

[0072] That is, the reason for this is that, when the block **230** is moved downwards, as shown in FIG. 4, the first support portion **271** must push the lead while being located on the upper surface of

the lead, and the first cutter **251** must be moved downwards more than the first support portion **271** in order to cut the lead.

[0073] At this time, the distance between the vertical surface of the seating die **211** and one side surface of the first cutter **251** that faces the vertical surface of the seating die **211** may be greater than 0 mm to less than 0.05 mm. This configuration is provided to prevent friction therebetween and to accurately cut the lead.

[0074] Next, a battery cell processing method using the processing apparatus described above will be described.

[0075] The battery cell processing method according to the present invention includes a first step of placing a battery cell to be processed on the upper surface of the seating die **211** of the base portion **210**, a second step of moving the seating die **211** forwards, a third step of moving the block portion **230** downwards in the vertical direction to cut a negative electrode lead **120** and a positive electrode lead **130** to a predetermined length and at the same time to form one or more embossed portions **121** in the negative electrode lead **120**, and a fourth step of separating the processed battery cell from the seating die **211**. The above steps are repeatedly performed.

[0076] In particular, when one edge of each of the cutters **250**, which corresponds to the edge of a knife, becomes dull as the result of processing a predetermined number of battery cells, the cutters **250** are separated from the block portion **230** and are then rotated 180 degrees clockwise or counterclockwise in order to cut leads using the other edges.

[0077] Meanwhile, in the case in which both of the opposite edges have been used, the bottom portion **251(c)** of each of the cutters **250**, each of which is made of a metal material, may be ground or cut by a predetermined thickness such that the front portion **251(a)** and the bottom portion **251(c)** form a right angle therebetween and the rear portion **251(b)** and the bottom portion **251(c)** form a right angle therebetween, whereby the cutters may be reused.

[0078] Although the specific details of the present invention have been described in detail, those skilled in the art will appreciate that the detailed description thereof discloses only preferred embodiments of the present invention and thus does not limit the scope of the present invention. Accordingly, those skilled in the art will appreciate that various changes and modifications are possible, without departing from the category and the technical idea of the present invention, and it will be obvious that such changes and modifications fall within the scope of the appended claims.

DESCRIPTION OF REFERENCE SYMBOLS

[0079] **100**: Battery cell [0080] **110**: Cell case [0081] **120**: Negative electrode lead **121**: Embossed portion [0082] **130**: Positive electrode lead [0083] **200**: Processing apparatus [0084] **210**: Base portion [0085] **211**: Seating die [0086] **220**: Vertical fixing portion [0087] **230**: Block portion [0088] **240**: First connection portion [0089] **250**: Cutter [0090] **251**: First cutter [0091] **251(a)**: Front portion **251(b)**: Rear portion [0092] **251(c)**: Bottom portion **251(d)**: Edge portion [0093] **251(e)**: Side portion **251(f)**: Extension portion [0094] **252**: Second cutter [0095] **260**: Second connection portion [0096] **261**: 2' connection portion **262**: 2'' connection portion [0097] **270**: Support portion [0098] **271**: First support portion **271(a)**: Protrusion [0099] **272**: Second support portion [0100] **280**: Third connection portion [0101] **281**: 3' connection portion **282**: 3'' connection portion

Claims

1. A battery cell of a secondary battery, the battery cell comprising: an electrode assembly; a cell case accommodating the electrode assembly; and a pair of electrode leads protruding outwards from the cell case, wherein the pair of electrode leads include a first electrode lead including a material that is the same as a material of a plate configured to connect the battery cell to another battery cell in series or parallel; and wherein the pair of electrode leads include a second electrode lead including a material that is different from the material of the first electrode lead.

2. The battery cell according to claim 1, wherein the first electrode lead includes one or more embossed portions, and the second electrode lead has no embossed portion.
 3. The battery cell according to claim 2, wherein the material of the first electrode lead includes nickel, and the material of the second electrode lead includes aluminum.
 4. The battery cell according to claim 2, wherein the material of the first electrode lead and the material of the plate include nickel, and wherein the one or more embossed portions of the first electrode lead and the material of the first electrode lead and the material of the plate including nickel avoids welding defects due to weak-welding or non-welding.
 5. The battery cell according to claim 2, wherein the first electrode lead is a negative electrode lead, and the second electrode lead is a positive electrode lead.
 6. The battery cell according to claim 1, wherein the cell case is configured as a laminate sheet structure including an inner layer, a metal layer, and an outer layer.
 7. The battery cell according to claim 6, wherein the inner layer is disposed in direct contact with the electrode assembly, and has insulation properties and resistance to an electrolytic solution.
 8. The battery cell according to claim 6, wherein the inner layer includes a material selected from among polypropylene, polyethylene, polyethylene-acrylic acid, polybutylene, a polyurethane resin, and a polyimide resin.
 9. The battery cell according to claim 6, wherein the metal layer is disposed to abut the inner layer, and is configured to prevent moisture or gas from permeating into the battery cell from an outside, and wherein the metal layer includes aluminum.
 10. The battery cell according to claim 6, wherein the outer layer is disposed to abut the metal layer on a side opposite to the inner layer, and is configured to have tensile strength, resistance to moisture permeation, and resistance to air transmission, heat resistance and chemical resistance, and wherein the outer layer includes nylon or polyethylene terephthalate.
 11. The battery cell according to claim 1, wherein the electrode assembly includes one of a jelly-roll type electrode assembly, a stacked type electrode assembly, a stacked and folded type electrode assembly, and a laminated and stacked type electrode assembly.
 12. The battery cell according to claim 1, wherein the first electrode lead and the second electrode lead are exposed out of the cell case in a state in which a first electrode tab and a second electrode tab are electrically connected respectively to the first electrode lead and the second electrode lead.
 13. The battery cell according to claim 1, wherein the first electrode lead and the second electrode lead are directly connected to the electrode assembly without connection to a first electrode tab and a second electrode tab.
 14. The battery cell according to claim 1, wherein the first electrode lead and the second electrode lead are not bent, but extend straight from the cell case.
 15. The battery cell according to claim 2, wherein the one or more embossed portions are hemispherical.
 16. The battery cell according to claim 2, wherein, when two or more embossed portions are formed in the first electrode lead, the two or more embossed portions are aligned in a width direction of the first electrode lead.
 17. The battery cell according to claim 1, wherein the first electrode tab and the second electrode tab protrude from the cell case in a same direction.
 18. The battery cell according to claim 1, wherein a melting point of the first electrode tab is greater than a melting point of the second electrode tab.
 19. The battery cell according to claim 1, wherein the cell case is rectangular.
 20. The battery cell according to claim 1, wherein the one or more embossed portions of the first electrode tab are formed adjacent to an edge of the first electrode tab that is distal from the cell case.
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